Do Now · retrieval

Noughts & Crosses · Part 2 — Discover iteration

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

In a zero-indexed C# array, which element does array[2] access?

Q2

Name the three parts of a C# for loop header used to visit every array element.

Q3

In a C# 2D array organised as rows and columns, what does GetLength(0) return?

Q4

If a board has 3 rows and 3 columns, how many values does it contain?

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Notice the repeated statements — Look at the three explicit row-printing instructions. The operation is the same; only the row index changes: 0, 1, 2 . — CODE:

```
Console.WriteLine(board[0,0] + " | " + board[0,1] + " | " + board[0,2]);
Console.WriteLine(board[1,0] + " | " + board[1,1] + " | " + board[1,2]);
Console.WriteLine(board[2,0] + " | " + board[2,1] + " | " + board[2,2]);
```

Core idea 2

Replace the rows with one loop — The first loop repeats the same row-print statement. row changes automatically; each repetition chooses a different row. — CODE:

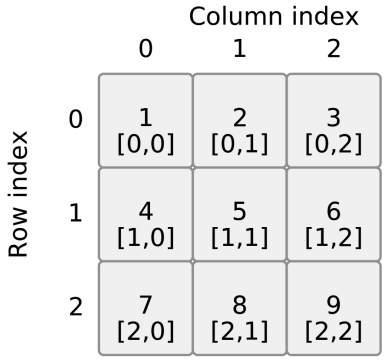
```
for (int row = 0; row < board.GetLength(0); row++)
{
Console.WriteLine(board[row,0] + " | " +
board[row,1] + " | " +
board[row,2]);
}
```

Core idea 3

Spot the second repetition — Inside each row, we still refer to columns 0, 1 and 2. A second loop can move across the columns of one row. — CODE:

```
for (int col = 0; col < board.GetLength(1); col++)
{
Console.Write(board[0,col]);
if (col < board.GetLength(1)-1)
Console.Write(" | ");
}
Console.WriteLine();
```

Key visual · complete concept



New Learning · applying the ideas

Core idea 4

Put one loop inside the other — Outer loop = down the rows. Inner loop = across the columns. This is called nested iteration . — CODE:

```
for (int row = 0; row < board.GetLength(0); row++)
{
for (int col = 0; col < board.GetLength(1); col++)
{
Console.Write(board[row,col]);
if (col < board.GetLength(1)-1)
Console.Write(" | ");
}
Console.WriteLine();
}
```

Core idea 5

Make it reusable — GetLength(0) gives the row count; GetLength(1) gives the column count. Place the loops in DrawBoard() , then call it whenever the display needs refreshing. — CODE:

```
DrawBoard(); // called from Main once the board exists
```

Vocabulary · use the correct technical terms

2D array — An array with two indexes.

row — A horizontal line of values.

column — A vertical line of values.

index — A zero-based position.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

Problem

From three statements to one loop, then two

Follow the worked method

1. Start with the old solution — CODE:
Console.WriteLine(board[0,0] + " | " + board[0,1] + " | " + board[0,2]);
Console.WriteLine(board[1,0] + " | " + board[1,1] + " | " + board[1,2]);
Console.WriteLine(board[2,0] + " | " + board[2,1] + " | " + board[2,2]); — It works. But it repeats the same pattern three times.

2. Replace the repeated columns — CODE:
for (int col = 0; col < board.GetLength(1); col++)
{
Console.Write(board[0,col]);
if (col < board.GetLength(1)-1)
Console.Write(" | ");
}
Console.WriteLine(); — Now each column is selected by col . After the third column, finish the row.

3. Refine into DrawBoard() — CODE:
public static void DrawBoard()
{
for (int row = 0; row < board.GetLength(0); row++)
{
for (int col = 0; col < board.GetLength(1); col++)
{
Console.Write(board[row,col]);
if (col < board.GetLength(1)-1)
Console.Write(" | ");
}
Console.WriteLine();
}
} — This method assumes board is stored at class level. Call DrawBoard() from Main() . You can reuse it after every move in Lesson 015.

Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Write or predict

Study the original three row-print statements. Which part remains identical and which part changes?

CODE:

```
Console.WriteLine(board[0,0] + " | " + board[0,1] + " | " + board[0,2]);
Console.WriteLine(board[1,0] + " | " + board[1,1] + " | " + board[1,2]);
Console.WriteLine(board[2,0] + " | " + board[2,1] + " | " + board[2,2]);
```

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Start with what worked

Use your three row-print statements from Lesson 014. Run them unchanged.

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 2

Build DrawBoard()
Combine your two loops to visit all nine squares and put them inside DrawBoard() . Keep board accessible from the method and call it in Main() .

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

board.GetLength(1) returns the number of columns in the board.

Use key vocabulary from memory

1. Explain 2D array in your own words.

2. Explain row in your own words.

3. Explain column in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.